TANMAY ADHIKARI

Applied AI/ML Engineer | Generative AI | Agentic Systems | Computer Vision

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Professional Summary

Applied AI/ML Engineer with 1.5+ years building production Generative AI, RAG, agentic, and Computer Vision systems at OneStep Greener and its Fortune 500 logistics sister company, TCI — converted from intern to full-time within 4 months. Delivered sub-2-second RAG retrieval, 95% CV detection accuracy, and up to 25% cost reduction through LangChain, LLM APIs, MCP, and YOLOv8-based pipelines shipped end-to-end from prototype to production.

Technical Skills

AI/ML: RAG, LangChain, LangGraph, MCP, OpenAI/Gemini/Qwen3 APIs, Prompt Engineering, Vector Search (FAISS, ChromaDB), BM25, NLP, YOLOv8, ONNX Runtime, ArcFace

Engineering: Python, SQL, FastAPI, Flask, React 18, MySQL, MongoDB, SQLite, Git, REST APIs, SSE, pytest, Docker-ready deployment

Analytics: Linear Programming, K-Means Clustering, Route Optimization, Measurable Impact Tracking

Professional Experience

Applied AI/ML Engineer | **OneStep Greener (TCI Group)** | *Nov 2025 – Present*

Converted from intern to full-time within 4 months; building AI systems deployed across OneStep Greener and sister company TCI

- Cut document search time to under 2 seconds for 500+ employees by architecting an enterprise RAG platform with semantic retrieval across multi-document repositories, deployed group-wide at sister company TCI
- Eliminated manual 24/7 dock monitoring at TCI logistics hubs by building a real-time computer vision automation system using edge inference on live RTSP streams
- Achieved 95%+ classification accuracy on waste segmentation by leading a CircularNet-aligned CV initiative processing live conveyor-belt feeds across recycling operations in collaboration with **Google**
- Improved model iteration speed by designing structured, bounding-box-annotated datasets feeding continuous CV model retraining

AI/ML Intern | **OneStep Greener (TCI Group)** | *Jul 2025 – Nov 2025*

- Increased fleet utilization by 20% by engineering driver-allocation logic using vehicle capacity and estimated waste volume data
- Optimized citywide waste-collection routing for Delhi by applying K-Means clustering with the Google Distance Matrix API

GenAI & ML Intern | **Genpact** | *Jan 2025 – Jun 2025*

- Reduced delivery costs by up to 15% by building last-mile route optimization models for Fortune 500 client operations

Key Projects

Conversational Data Analytics Platform | *Python, FastAPI, React 18, MCP, MySQL, BM25, SSE*

- Enabled non-technical staff to query a live 67-table operations database in plain English by architecting a hybrid RAG pipeline that routes each query across three modes (document, SQL, hybrid) via a read-only MCP server — eliminating LLM schema hallucination
- Blocked 100% of prompt-injection and mutation attempts by engineering a defense-in-depth SQL safety layer: allow/deny-list validation, read-only transactions with 15s timeouts, and hard row caps
- Cut LLM cost and latency by short-circuiting off-topic queries with zero-token rule-based guards and curated FAQ query recipes; shipped multi-provider Gemini fallback, SSE token streaming, and 109 automated tests with no live dependencies

Contactless Facial-Recognition Attendance System | *Python, Flask, React 18, ONNX Runtime, FAISS, MongoDB, SQLite*

- Achieved sub-second worker identification at industrial recycling sites by building a training-free SCRFD + ArcFace recognition pipeline on ONNX Runtime with in-memory FAISS cosine-similarity search over multi-vector profiles
- Improved recognition reliability across variable real-world lighting by designing an adaptive matching algorithm that fuses per-user vector scores with detection confidence and dynamically adjusts thresholds by brightness, sharpness, and pose quality
- Delivered a unified single-process Flask + React 18 deployment behind nginx with PIN-protected enrollment, burst-capture registration, live attendance analytics with CSV export, and a dual SQLite + MongoDB persistence layer with idempotent schema migration

Education

B.Tech, Computer Science Engineering — Graphic Era Hill University, Dehradun | *2021 – 2025*